

specialists, market makers were not privileged to the entire market buy-sell imbalance, as they only had access to their central order book consisting of their customer orders. The privilege differences provided some difficulties at times in establishing fair value market prices due to the fragmentation of orders. Still, the competitive system played a large role in determining fair market prices.

Times have changed, and the underlying trading process is far from the simple process of yesteryear. There are more than thirteen displayed trading venues including exchanges and alternative trading systems (ATSs) and from thirty to forty electronic crossing networks (ECNs) which are known as electronic communication networks. To make matters more difficult, the rules of engagement for each venue differ with several different pricing and queuing models. For example, maker-taker, inverted, commission based, price-time, price-size, etc.

To keep abreast of the rapidly changing market structure, many funds have created an in-house market microstructure analyst role and some have hired a team of analysts. In order to develop the most efficient and profitable trading algorithms and systems, market participants need to fully understand the inner workings of each venue and how it affects their orders, and ultimately their bottom line.

We provide an overview of the different venues and provide empirical data relating to intraday trading patterns such as spreads, volumes, volatility, and trading stability. The chapter concludes with a discussion of the market microstructure surrounding special event days, with special attention paid to the “Flash Crash” of May 6, 2010 and what really happened during the day.

In the remainder of this chapter, we describe the current state of the equity markets and review some of the important market microstructure research papers that have not only provided the groundwork for the evolution of exchanges, but also provided a foundation for trading algorithms, high frequency trading, and ultimately improved stock selection and portfolio construction.

Readers are encouraged to study these research works and empirical findings as if it were a final exam.

MARKET MICROSTRUCTURE LITERATURE

Academia has been performing market microstructure research since the 1970s. And while the research has expanded immensely, the current work